NATIONAL INSTITUTE OF BUSINESS MANAGEMENT

In collaboration with COVENTRY UNIVERSITY

High National Diploma in Data Science

Information Retrieval Project 1 (Individual Project)

Web Crawling and Indexing - (15 marks of the final module)

Students will develop an information retrieval system that crawls a website starting from a seed page, stores at least 1000 pages, and Index them using an Inverted Index in Python.

The project is divided into 2 main sections, each with specific tasks and deliverables. Each section will be graded separately.

Section 1: Crawling (60 marks)

Task: Develop a web crawler that starts from a seed page and collects at least 500 unique web pages from the same domain.

Requirements:

1. Seed Page: Start crawling from a given URL (seed page).
2. Page Limit: Store at least 500 unique pages. (DO not exceed 1000 pages.)
3. Implement a delay between requests to avoid overloading the server.
4. Error Handling: Handle exceptions and retry failed requests.
5. Storage: Save the content of each page in a local directory. (Do not save them as HTML and save them as HTML pages.)

Section 2: Indexing (40 marks)

Task: Create an index of the crawled web pages.

Requirements:

1. Tokenization: Break down the text content of each page into tokens (words).
2. Normalization: Convert tokens to lowercase, remove punctuation, and apply stemming/lemmatization.
3. Inverted Index: Build an inverted index mapping terms to document IDs.
4. Storage: Save the index in a suitable data structure (e.g., dictionary) and store it in a file (JSON).

Submission Guidelines

Submit a single Python script with the following regions separated by comments:

- Web Crawler
- Indexing

Additionally, prepare below during the presentation:

- Directory containing the stored web pages.
- Inverted index file.

Notes:

Prohibition of AI Tools: The use of AI language models, such as ChatGPT, Gemini, or similar tools, is strictly prohibited for this project. This includes generating code, ideas, or any other content. Your work must be original and independently created. (You are allowed to search the web and get codes to get an idea. But you must modify the code according to your need,)

Domain Selection: Each Individual must select a different domain for their project. This will ensure diversity in the topics covered and allow for a broader understanding of Information Retrieval concepts. Please discuss and finalize your domain choice with me and the group before proceeding. At the later stage you will be merged with your group members for Project 2 to create an Information Retrieval system.

Code Questions: During the project assessment, you will be expected to answer questions regarding your code. The assessment of your answers will be based on your understanding and explanation of your implementation.

***** End of the Instructions *****